

① Introduction to Statistics for Data Science.

"STATISTICS"

↳ Science of $\left. \begin{array}{l} \rightarrow \text{collecting} \\ \rightarrow \text{Analyzing} \\ \rightarrow \text{Interpreting} \\ \rightarrow \text{Presenting} \end{array} \right\} \text{data}$

→ Every ML model depends on Statistical foundation.

◦ Types: Descriptive & Inferential

↓
Summarize & describe data
eg: Avg classroom marks

↓
Draws conclusions about 'P' on a 'S'.
eg: check 50 items from batch of 10,000 for defect rate.

◦ Uses of Statistics in Data Science:

- EDA (Data Understanding)
- Decision Making
- Prediction & Modeling
- Risk & Uncertainty estimation.

② Descriptive Statistics:

→ [Summarize & describe]

eg:- Dataset of scores.

→ [68, 72, 75, 75, 78, 80, 85, 90, 95]

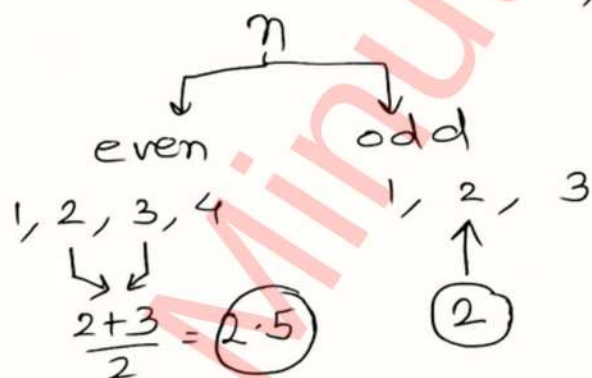
① Central Tendency

- Mean : Avg (balances all values)
- Median : Middle value
- Mode : Most frequent value.

$$\begin{aligned}\rightarrow \text{Mean} &= \mu = \frac{\text{Sum}}{\text{Total}} = \frac{1}{n} \sum_{i=1}^n x_i \\ &= \frac{718}{9} = \underline{79.77}\end{aligned}$$

→ Median ⇒ Sorted the values (already)

68, 72, 75, 75, 78, 80, 85, 90, 95



⇒ $n = 9$ (odd)

$$\begin{aligned}\text{Middle index} &= \frac{n+1}{2} \\ &= \frac{10}{2} = 5\end{aligned}$$

So Median = 78

→ Mode : most frequent value = 75

• Types of Data

- ↳ Categorical
- ↳ Quantitative
- ↳ Time series.

⇒ Categorical

- ↳ Nominal (eg: color of shirt)
- ↳ Ordinal. (Size of shirt)

⇒ Quantitative

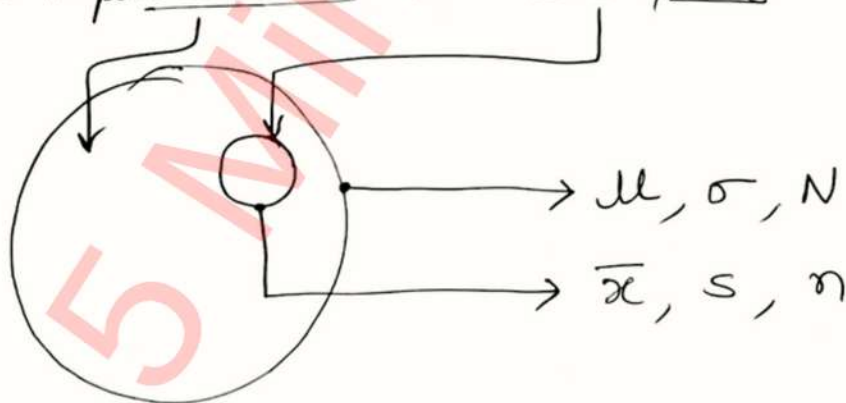
- ↳ Discrete (eg: no. of children)
- ↳ Continuous (height of children)

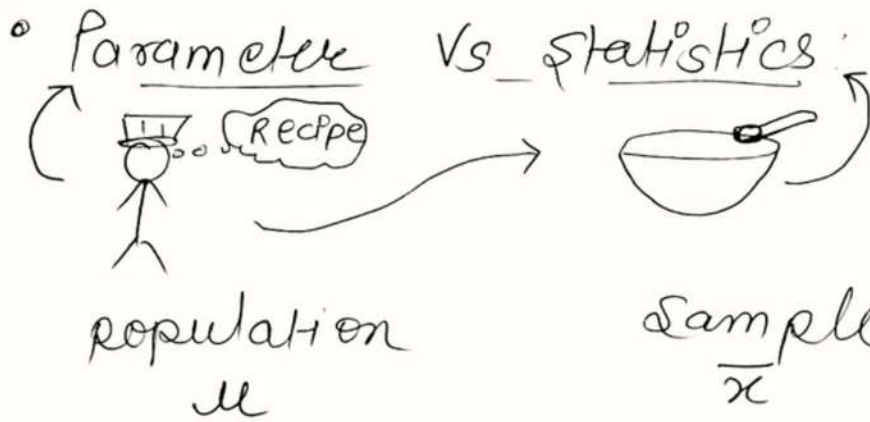
⇒ Time Series

- ↳ Monthly sale growth of business.

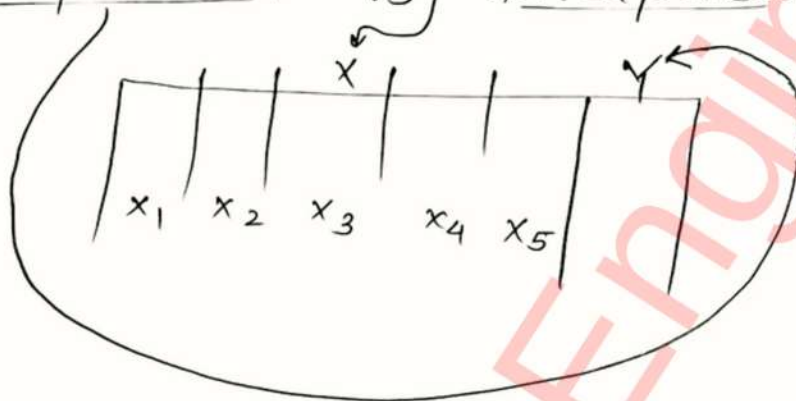


• Population Vs Sample:



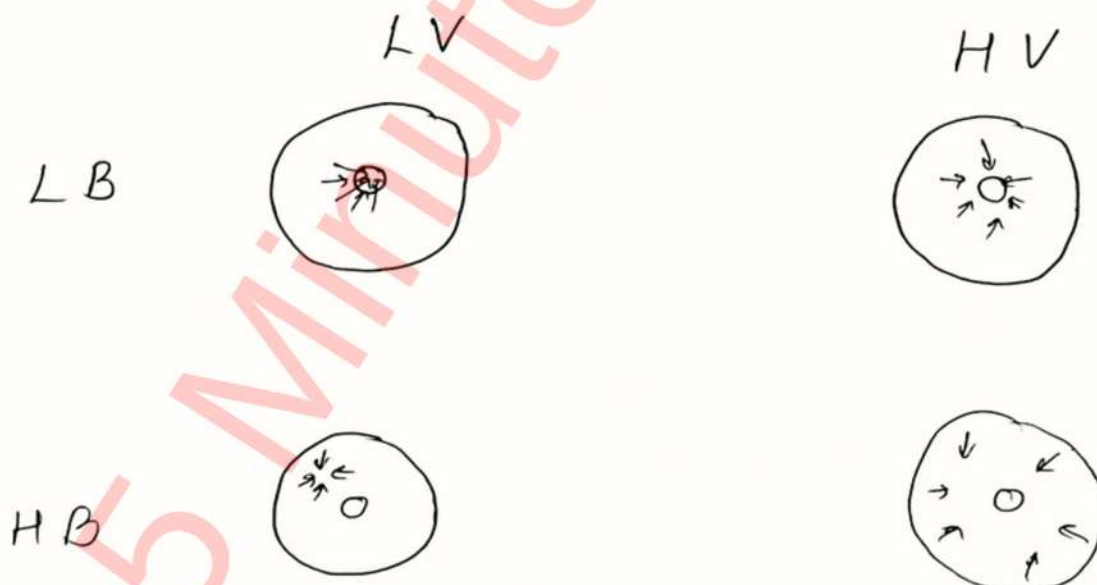


Dependent Vs Independent variables:



$$Y = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + b_4 x_4 + b_5 x_5$$

Bias & Variance



- ② Dispersion: It quantifies spread/variability around centre
- Range: Max - Min
 - Variance / SD: Avg squared deviation
 - Coef. of Variation scales SD by mean. (CV)

Imp: High spread $\Rightarrow$ Uncertain prediction
CV compares variability across features with different units.

eg:- $\mu = 79.77$, $n = 9$

$$\begin{aligned}\rightarrow \text{Range} &= \text{max} - \text{min} \\ &= 95 - 68 \\ &= 27\end{aligned}$$

$$\begin{aligned}\rightarrow \text{Variance} &= \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 \\ &= \frac{611.55}{9} = 67.95\end{aligned}$$

$$\begin{aligned}\rightarrow \text{SD} &= \sqrt{\sigma^2} \\ &= \sigma = \sqrt{67.95} = 8.24\end{aligned}$$

$$\begin{aligned}\rightarrow \text{CV} &= \frac{\sigma}{\mu} = \frac{8.24}{79.77} = 0.1032 \\ &\approx 10.32\%\end{aligned}$$

③ Quartiles, IQR & Outliers

↳ splits the data in 4 parts.

→ $Q_1 = (25^{\text{th}} \text{ percentile})$

→ $Q_2 = (50^{\text{th}} \text{ percentile})$ [Median]

→ $Q_3 = (75^{\text{th}} \text{ percentile})$

→ $Q_0 \text{ \& } Q_4 = \text{min \& max.}$

eg:- 68, 72, 75, 75, 78, 80, 85, 90, 95

• $Q_1 \rightarrow$ Lower half: [68, 72, 75, 75]

$$\downarrow$$
$$\frac{72 + 75}{2} = 73.5$$

• $Q_3 \rightarrow$ Upper half: [80, 85, 90, 95]

$$\downarrow$$
$$\frac{85 + 90}{2} = 87.5$$

$$\begin{aligned} \bullet \text{ IQR} &= Q_3 - Q_1 \\ &= 87.5 - 73.5 \\ &= 14 \end{aligned}$$

• Outlier fences

$$\begin{aligned} \rightarrow \text{Lower fence} &= Q_1 - 1.5 \times \text{IQR} \\ &= 73.5 - 21 = \underline{52.5} \end{aligned}$$

$$\begin{aligned} \rightarrow \text{Upper fence} &= Q_3 + 1.5 \times \text{IQR} \\ &= 87.5 + 21 = \underline{108.5} \end{aligned}$$

④ Shape

- Skewness: Measures Asymmetry
- Kurtosis: Heavy tail vs Narrow.

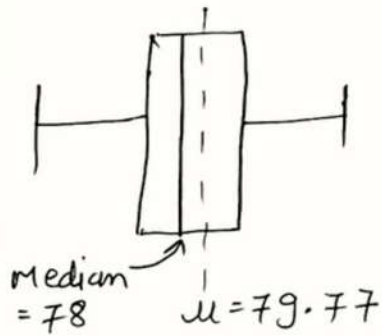
eg:- Pearson's 2nd skewness:

$$\text{Skewp} = \frac{3 (\mu - \text{Median})}{\sigma}$$

$$= \frac{3 (79.77 - 78)}{8.74}$$

$$= \frac{5.33}{8.74}$$

$$= 0.6100$$



$$\Rightarrow \text{Skewp} \approx 0.61 > 0 \text{ [+ve skew]}$$

+ve skew
(Right skewed)

$$\mu > \text{Median}$$

-ve skew
(Left skewed)

$$\mu < \text{Median}$$

$$\begin{aligned} \circ \text{Kurtosis} &= \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^4}{\left[\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 \right]^2} \\ &= \frac{4^{\text{th}} \text{ CM}}{(2^{\text{nd}} \text{ CM})^2} \end{aligned}$$

$$1) \mu = \frac{718}{9} = 79.77$$

2) Compute Deviation & Moments:

x_i	$x_i - \mu$	$(x_i - \mu)^2$	$(x_i - \mu)^4$
68	-11.77	138.64	19234.64

$$\Rightarrow \sum (x_i - \mu)^2 = 611.56$$

$$\Rightarrow \sum (x_i - \mu)^4 = 89308.82$$

3) Compute 2nd & 4th CMs.

$$\mu_2 = \frac{\sum (x_i - \mu)^2}{n} = \frac{611.56}{9} = 67.95$$

$$\mu_4 = \frac{\sum (x_i - \mu)^4}{n} = \frac{89308.82}{9} = 9923.20$$

4) Compute kurtosis

$$\hookrightarrow \frac{\mu_4}{(\mu_2)^2} = \frac{9923.20}{(67.95)^2}$$

$$= 2.1491$$

Since: $\beta_2 < 3$; It's platykurtic
; Tail is lighter
(few extreme outliers)

⑤ Standardization - 'Z' scores

↳ z score measures how many SDs a value is away from mean: $z = \frac{x - \mu}{\sigma}$

$z > 0$: value is above mean

$z < 0$: value is below mean.

eg:- [68, 72, 75, 75, 78, 80, 85, 90, 95]

$$\rightarrow \mu = 79.77, \sigma = 8.24$$

$$\begin{aligned} \therefore \text{for } x = 95: \quad z &= \frac{95 - 79.77}{8.24} \\ &= \underline{1.84} \end{aligned}$$

$$\begin{aligned} \therefore \text{for } x = 68: \quad z &= \frac{68 - 79.77}{8.24} \\ &= \underline{-1.42} \end{aligned}$$

Both lie within $\pm 2 \sigma$ (SDs) [Not outliers]

if $(|z| > 3 \rightarrow \text{unusual})$

◦ Frequency Tables:

Apple	- 2
Bg	- 5
xyz	- 10
abc	- 3

eg:- $A = 3, B = 2, C = 3$

Total items = 8

Relative freq = $\frac{3}{8}, \frac{2}{8}, \frac{3}{8}$

$\Rightarrow$	<u>value</u>	<u>count</u>	<u>%</u>
	A	3	37.5%
	B	2	25%
	C	3	37.5%

◦ Cross-tabulation [Contingency Table]

eg:- Gender : [M, F, F, M, M, F, M, F]

Sports : [C, C, F, F, C, F, C, C]

$\Rightarrow 4 \times 5 \downarrow$

<u>Gender</u>	<u>Cricket</u>	<u>Football</u>	<u>Total</u>
M	3	1	4
F	2	2	4
<u>Total</u>	5	3	8

• Probability Theory

→ Basics of Probability:

$$P(A) = \frac{\text{favorable outcomes}}{\text{Total outcomes}}$$

Diagram showing the numerator of the probability formula branching into two possibilities:

- Not possible (0)
- possible (1) ✓

eg: Roll a fair die → $P(\text{getting } 4)$

⇒ Total: $\{1, 2, 3, 4, 5, 6\} \rightarrow 6$

⇒ favorable: $\{4\} \rightarrow 1$

$$\therefore P(4) = 1/6 = 0.1667.$$

→ Mutually Exclusive & Independent Events

↓
 $P(A \cap B) = 0$

[can't happen together]

↓
 $P(A \cap B) = P(A) P(B)$

[doesn't affect each other]

eg: 52 cards of deck.

$$P(\text{King}) = 4/52 = 1/13$$

$$P(\text{Queen}) = 4/52 = 1/13$$

$$P(K \cap Q) = 0$$

eg: Flip coin & roll a die

$$\text{Coin flip head} = 1/2$$

$$\text{Die shows} = \frac{1}{6}$$

$$P(H \cap 6) = \frac{1}{2} \times \frac{1}{6}$$

$$= \frac{1}{12}$$

→ Conditional Probability

$$\hookrightarrow P(A/B) = \frac{P(A \cap B)}{P(B)}$$

eg: Event $A = \text{King}$.

$B = \text{face card (J, Q, K)}$

$$\therefore P(A) = 4/52$$

$$P(B) = 12/52$$

$$P(A \cap B) = 4/52$$

$$\begin{aligned}\therefore P(A/B) &= (4/52) \div (12/52) \\ &= 4/12 \\ &= \underline{1/3}\end{aligned}$$

→ Total probability:

If $(B_1, B_2, \dots, B_n)$ is a partition:

$$P(A) = \sum P(A/B_i) P(B_i)$$

eg:- Student passes an exam.

70% studied, 30% didn't

= If studied $\rightarrow P(\text{pass}) = 0.9$

= If didn't study $\rightarrow P(\text{pass}) = 0.4$.

$$\begin{aligned}\therefore P(\text{pass}) &= 0.9 \times 0.7 + 0.4 \times 0.3 \\ &= 0.63 + 0.12 = 0.75\end{aligned}$$

→ Bayes Theorem

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

eg: • 1% of population

has disease $\rightarrow P(D) = 0.01$

• Test detects 'D' correctly $\rightarrow P(\text{Pos}/D) = 0.95$

• false positive rate $\rightarrow 5\% \rightarrow P(\text{Pos}/\sim D) = 0.05$

To find $P(D/\text{pos})$

$\Rightarrow$ 1) Total $P(\text{pos})$:

$$= P(\text{Pos}/D) P(D) + P(\text{Pos}/\sim D) P(\sim D)$$

$$= 0.95 \times 0.01 + 0.05 \times 0.99$$

$$= 0.0095 + 0.0495$$

$$= 0.059$$

$\Rightarrow$ 2) Bayes:

$$P(D/\text{Pos}) = \frac{(0.95 \times 0.01)}{0.059}$$

$$= 0.161$$

$$\approx 16.1\%$$

→ Random Variables:

↳ eg:- Rolling a die,

The number showing (1-6) values
is a RV

Types:

- Discrete (finite or countable)
- Continuous (Infinite)

Key functions:

- PMF (Probability Mass fn)
- PDF (Probability density fn)
- CDF (Cumulative distribution fn)

Numerical (DRV)

↳ Toss a coin twice.

X = no. of heads.

① → Sample space = $\{HH, HT, TH, TT\}$

② → Map outcomes = values of X :

↓
 $HH \rightarrow 2, HT \rightarrow 1, TH \rightarrow 1, TT \rightarrow 0$

③ → Count probabilities:

$$P(X=0) = 1/4$$

$$P(X=1) = 2/4$$

$$P(X=2) = 1/4$$

∴ PMF:

X	<u>0</u>	<u>1</u>	<u>2</u>
$P(X=x)$	0.25	0.5	0.25

Numerical (CRV)

A bus arrives uniformly at random betⁿ 0 & 10 minutes.

Let x = waiting time.

① PDF = $f(x) = 1/10$ for $0 \leq x \leq 10$

② Probabilities = area.

$$\Rightarrow P(2 \leq x \leq 5) = \int_2^5 (1/10) dx$$

$$\Rightarrow (1/10)(5-2) = 3/10 = 0.3$$

$\therefore$ 30% chance you wait betⁿ 2-5 minutes

③ CDF : for $0 \leq x \leq 10 \rightarrow f(x) = x/10$
eg: $P(x \leq 7) = f(7) = 7/10 = 0.7$

$\Rightarrow$ Expectation & Moments

$\downarrow$
(mean / Avg)
long run.

$\left\{ \begin{array}{l} \rightarrow 2^{\text{nd}} \text{ Moment : Variance} \\ \rightarrow 3^{\text{rd}} \text{ Moment : Skewness} \\ \rightarrow 4^{\text{th}} \text{ Moment : Kurtosis} \end{array} \right.$

eg: Roll a fair die:

① PMF : each value 1-6 $\rightarrow P = 1/6$

② Expectation: $E[x] = (1+2+3+4+5+6)/6 = 3.5$

③ $E[x^2] = 91/6 = 15.16$

④ $\text{Var}(x) = E[x^2] - (E[x])^2 = 15.16 - 12.25 = 2.917$

⑤ $\text{std dev} = \sqrt{2.917} = 1.71$

• Probability Distribution

① Discrete Distribution

↳ Bernoulli . D

eg: flipping a coin

↳ p is either 0 or 1
(Binary classification)

$$\text{PMF : } P(X=1) = p, P(X=0) = 1-p$$

$$\text{eg: } P(H) = 0.7$$

$$\therefore P(X=1) = 0.7$$

$$P(X=0) = 0.3$$

$$E[X] = p = 0.7$$

$$\text{Var}[X] = p(1-p) = 0.7 \times 0.3 = 0.21$$

→ Binomial . D

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

eg: - $n = 5$ tosses

$$p = 0.5, P(X=2) = ?$$

$$P(X=2) = \binom{5}{2} (0.5)^2 (0.5)^3$$

$$= 10 \times 0.25 \times 0.125 = 0.3125$$

→ Geometric D

eg: How many time you roll a die
until you get 6
⇒ first success after
'k' trials.

$$\therefore \text{PMF} = P(X=k) = (1-p)^{k-1} p.$$

eg: $p=0.2$, $P(X=3)$? (Success on 3rd trial)

$$P(X=3) = (0.8)^2 (0.2)$$
$$= 0.128$$

→ Poisson D

eg: no. of emails arriving per hour.

→ Events occur at constant rate λ per interval

$$\text{PMF: } P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

eg: $\lambda = 3$ mails/hours, find $P(X=2)$

$$P(X=2) = e^{-3} \frac{3^2}{2!}$$
$$= \underline{0.224}$$

→ Multinomial D

$$\text{PMF} = P(x_1, x_2, \dots, x_k) = \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}$$

eg: Roll die $n=3$ times.

$$P(1 \text{ once, } 2 \text{ once, } 3 \text{ once}) = \frac{3!}{1! 1! 1!} \left(\frac{1}{6}\right)^3$$
$$= 0.0278$$

② Continuous Distribution

→ Uniform D: Randomly pick no. from $0 \rightarrow 10$
all uniformly/equally likely.

$$\text{PDF: } f(x) = \frac{1}{(b-a)} \text{ for } a \leq x \leq b$$

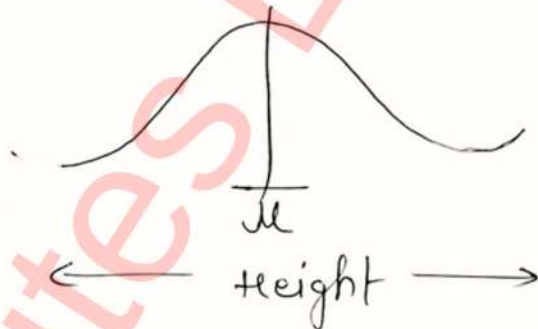
$$\text{CDF: } f(x) = (x-a)/(b-a)$$

eg:- $X \sim U(0, 10)$.

Probability $2 \leq x \leq 5$?

$$= (5-2)/(10-0) = \frac{3}{10} = 0.3.$$

→ Normal (Gaussian) D



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

eg: $x \sim N(\mu = 175 \text{ cm}, \sigma = 10 \text{ cm})$

Q: $P(x < 160)$

$$\rightarrow z = \frac{x - \mu}{\sigma} = \frac{-15}{10} = -1.5$$

for $P(z < -1.5) = 0.0668$

$\therefore 6.7\%$

eg: C1: $\mu = 50$, $\sigma = 10$
C2: $\mu = 60$, $\sigma = 5$

for $x = 55$

for C1:

$$f(55/\mu_1, \sigma_1) = \frac{1}{\sqrt{2\pi} (10)} e^{-\frac{(55-50)^2}{2(10)^2}}$$

$$= \underline{0.0352}$$

for C2:

$$f(55/\mu_2, \sigma_2) = \frac{1}{\sqrt{2\pi} (5)^2} e^{-\frac{(55-60)^2}{2(5)^2}}$$

$$= \underline{0.0484}$$

• Inferential Statistics:

↳ To infer / Conclude for P from S .

→ Sampling techniques:

① Simple Random Sampling:

↳ Lucky draw.

eg:- $P = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
($N=10$)

$n = 3$ samples randomly.

→ $\frac{3}{10} = 30\%$ chance.

$$\Rightarrow \{2, 5, 9\} \rightarrow \bar{x} = 16/3 = 5.33$$
$$\Rightarrow \mu = 55/10 = 5.5$$

Our estimate (5.33) is close to True mean.

② Stratified Sampling.

↳ maintain original ratio.

eg:- 100 emp. (70 Engineer, 30 Managers)

So for $n=10 \rightarrow \frac{70}{100} \times 10$ & $\frac{30}{100} \times 10$

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3

③ Cluster Sampling

↳ Randomly select entire clusters,
NOT Individuals.

eg:- $P = 1000$ schools in 10 cities.
(100 each).

We want $n = 200$.

So, we randomly choose any 2 cities.
to get 200 schools.

④ Systematic Sampling:

↳ first Random, then test/check/choose
every k^{th} value.

eg:- $P = \{1, 2, 3, \dots, 20\}$, $n = 5$.

$\therefore$ Sampling interval $= 20/5 = 4 = k$

If start random at 2 then,

$\{2, 6, 10, 14, 18\}$

$\bar{x} = 50/5 = 10$
 $\mu = 10.5$ } close.

• Estimation.

① Point Estimations:

↳ Single number estimate of μ .
(use $\bar{x}$ to estimate μ)

a) Sample μ ($\bar{x}$)

b) MLE (Maximum Likelihood Estimation)

eg:- Sample waiting time = $[0.2, 0.4, 1, 0.3, 0.6]$
 (t)
 $n = 5$

$$MLE = \frac{n}{\sum t_i}$$

$$\begin{aligned} \text{i) } 0.2 + 0.4 &= 0.6 \\ &+ 1.0 \\ &\hline 1.6 + 0.3 &= 1.9 \\ &+ 0.6 \\ &\hline 2.5 \end{aligned}$$

$$\text{ii) } \hat{\lambda} = \frac{5}{2.5} = 2 \text{ per hour.}$$

② Interval Estimation. & CI.

↳ Range of plausible values.



Parameter (Height) $\bar{x} = 175\text{cm}$; $SD = 20\text{cm}$

$$CI = 95\% ; \bar{x} \pm z \frac{s}{\sqrt{n}}$$

$$80\% \rightarrow 1.282$$

$$85\% \rightarrow 1.440$$

$$90\% \rightarrow 1.645$$

$$95\% \rightarrow 1.960$$

$$99\% \rightarrow 2.576$$

$$99.5\% \rightarrow 2.807$$

$$99.9\% \rightarrow 3.291$$

z values

$$\Rightarrow 175 \pm 1.960 \times \frac{20}{\sqrt{40}}$$

$$\Rightarrow 175 \pm 6.20$$

$$\downarrow$$
$$(175 - 6.20, 175 + 6.20)$$
$$L 168.8 \text{ cm}, 70, 181.2 \text{ cm}$$

• Chi-Square Test

eg:-

Watch Actor ↓	Y	N	Total (R)
M	140	44	184
F	178	38	216
Total (C)	318	82	400

$$\alpha = 0.05$$

$$df = (R-1) \times (C-1) = 1$$

H_0 : No Connection

H_1 : Full Connection.

→ Table value = 3.841

$$\chi^2_c = \sum \frac{(O_i - E_i)^2}{E_i}$$

$$E = \frac{RT \times CT}{T} = \frac{184 \times 318}{400} = 146$$

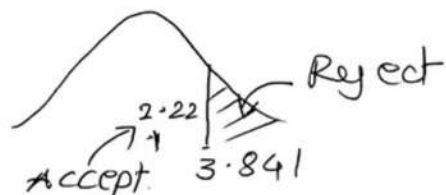
$$= \frac{184 \times 82}{400} = 38$$

$$= \frac{216 \times 318}{400} = 172$$

$$= \frac{216 \times 82}{400} = 44$$

Actor	Watch	O	E	$(O-E)^2$	$(O-E)^2/E$
M	Y	140	146	36	0.246
M	N	44	38	36	0.947
F	Y	178	172	36	0.209
F	N	38	44	36	0.818
					<u>2.220</u>

$$2.220 < 3.841$$



• Hypothesis Testing

eg1

$$H_0 : \mu = 50$$

$$H_1 : \mu \neq 50$$

$$\sigma = 20$$

$$n = 100$$

$$\bar{x} = 52$$

$$z = \frac{(\bar{x} - \mu)}{\sigma / \sqrt{n}}$$

$$= \frac{52 - 50}{20 / 10}$$

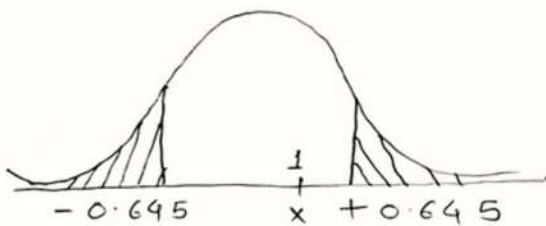
$$= \frac{2}{2}$$

$$\underline{z = 1}$$

$\alpha = 1\%$
(± 2.58)

$\alpha = 5\%$
(± 1.96)

$\alpha = 10\%$
(± 0.645)



$1 > +0.645$ Reject H_0 .

eg2

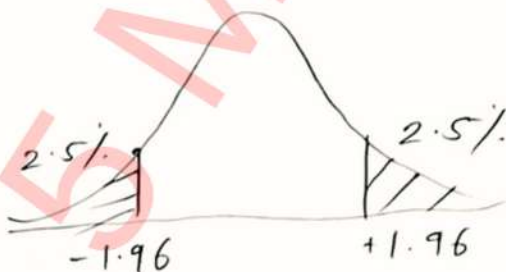
B1	B2
$\bar{x}_1 = 75$	$\bar{x}_2 = 73$
$\sigma_1 = 8$	$\sigma_2 = 10$
$n_1 = 60$	$n_2 = 100$
$H_0 : \mu_1 = \mu_2$	
$H_1 : \mu_1 \neq \mu_2$	
$\alpha = 5\%$	

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$= \frac{(75 - 73) - (0)}{\sqrt{\frac{(8)^2}{60} + \frac{(10)^2}{100}}}$$

$$= \frac{2}{1.437}$$

$$\underline{z = 1.391}$$



$1.391 < 1.96$ Accept H_0

◦ 2 Tail Vs 1 Tail

↓
 $\mu \neq 50$
 Not equal
 Non-directional

↓	↓
LTT	RTT
$\mu < 50$	$\mu > 50$
Smaller	higher
lower	more
Reduced	exceed
decreased	increased
Under	above.
less than	greater than.

◦ T-test

eg!

$$H_0: \mu \geq 60$$

$$H_1: \mu < 60 \text{ (LTT)}$$

$$\alpha = 5\%$$

$$n = 16$$

$$S = 40$$

$$\bar{x} = 70$$

$$\begin{aligned} df &= n-1 \\ &= 16-1 \\ &= 15 \end{aligned}$$

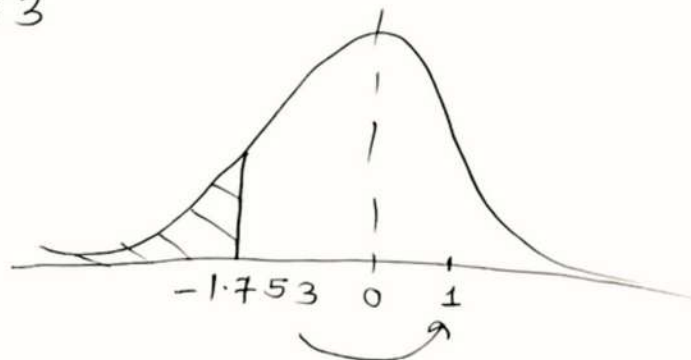
$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

$$= \frac{70 - 60}{40/\sqrt{16}}$$

$$= \frac{10}{10}$$

$$t = 1$$

$$\therefore t_c = -1.753$$



eg2: $H_0: \mu \leq 400$
 $H_1: \mu > 400$ (RTT)

$\alpha = 5\%$

$n = 12$

$S = 40$

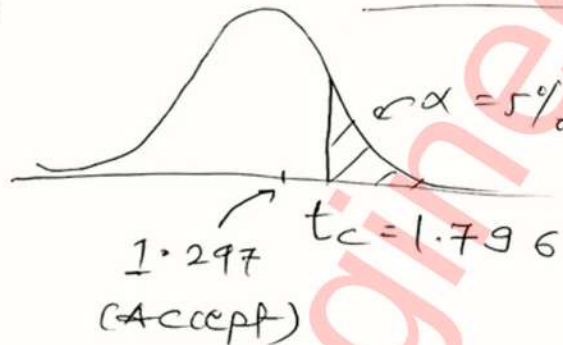
$\bar{x} = 415$

$df = 12 - 1$
 $= 11$

$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$

$= \frac{415 - 400}{40/\sqrt{12}}$

$t_{cal} = 1.297$



eg3: $H_0: \mu = 10$

$H_1: \mu \neq 10$ (2TT)

$n = 16$

$S = 8$

$\bar{x} = 20$

$\alpha = 5\%$

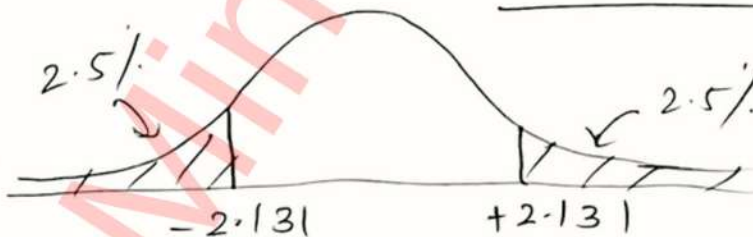
$df = 16 - 1$
 $= 15$

$t_c = \pm 2.131$

$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$

$= \frac{20 - 10}{8/\sqrt{16}}$

$= \frac{10}{2} = 5$



$|t_{cal}| > t_c$

$5 > 2.131$

(H_0 Reject)

• ANOVA (Analysis of Variance)

eg: 3 groups (each group has 3 nos).

→ GA : 3, 4, 5

→ GB : 4, 5, 6

→ GC : 8, 9, 10

N = 9 & k = 3
Total obs. no. of groups

① Group sum & mean

→ GA → sum = 12
→ mean = $12/3 = 4$

→ GB → sum = 15
→ mean = $15/3 = 5$

→ GC → sum = 27
→ mean = $27/3 = 9$

→ Total → sum = 54
→ mean = $54/9 = 6$

② SSB (Sum of Squares betn groups)

↳ formula: $\sum (n \times (\underbrace{G_m}_{\text{Group mean}} - \underbrace{G_{rm}}_{\text{Grand mean}})^2)$

→ for Group A

↳ $(4 - 6)^2 = (-2)^2 = 4$

→ $n \times 4$
 $= 3 \times 4 = 12$

→ for Group B

↳ $(5 - 6)^2 = (-1)^2 = 1$

→ $3 \times 1 = 3$

→ for Group C

↳ $(9 - 6)^2 = (3)^2 = 9$

→ $3 \times 9 = 27$

∴ $12 + 3 + 27 = 42 = \underline{\underline{SSB}}$

③ SSW (Sum of Squares within groups)

↳ $\sum \sum (x_{ij} - G_m)^2$

→ for Group A (mean = 4):

$(3 - 4)^2 = (-1)^2 = 1$

$(4 - 4)^2 = (0)^2 = 0$

$(5 - 4)^2 = (1)^2 = 1$

sum = $1 + 0 + 1 = 2$

for Group B : Sum of B = 2

for Group C : Sum of C = 2.

$$\therefore SSW = 2 + 2 + 2 = 6$$

④ Total = $SSB + SSW = 42 + 6$
variability = 48

⑤ $df_bet^n = k - 1 = 3 - 1 = 2$
 $df_within = N - k = 9 - 3 = 6$

⑥ Mean Squares & f statistics

$$\rightarrow MS_bet^n = SSB / df_bet^n = 42 / 2 = 21$$

$$\rightarrow MS_within = SSW / df_within = 6 / 6 = 1$$

$$\rightarrow F = \frac{MS_bet^n}{MS_within} = \frac{21}{1} = 21$$

⑦ Interpret & conclude.

$$F = 21 \gg 1$$

$\therefore$ At least one Group mean is different

[f-table for $df = (2, 6) = 5.14 > 21$ (reject)] (Group C = 9)

• Correlation $\sqrt{s}$ Covariance

$$\text{Cov}(X, Y) = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

↓

> 0 (together)

< 0 (opposite)

$$\frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

↓

+1 (true linear)

-1 (-ve linear)

0 (No linear)

eg: $n = 3$

X (study hours) = [1, 2, 3]

Y (test score) = [2, 4, 5]

$$\Rightarrow \bar{x} = 2$$

$$\bar{y} = 3.67$$

$$\sigma_X = 0.816$$

$$\sigma_Y = 1.247$$

$$\therefore \text{Cov}(X, Y) = \frac{3}{3} = 1 \text{ (increase together)}$$

$$\therefore r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{1}{0.816 \times 1.247} = 0.984$$

↓

(+ve linear
Relationship)